Business Engineering Recruitment Case Study

Process Description :

In this case study you will be creating a report about a logistics process relevant to one of our key business-unit, “Autohero”.
All autohero cars must go to a workshop in order to be inspected and refurbished.
First, the car is booked to Autohero, then it needs to be transported to a workshop. At any moment, the car can be ‘unbooked’ and is therefore withdrawn from this process.

The transport to workshops from branches can happen directly or via one of our logistics centers.

Explanation of the datasets provided :

For this case study you are provided with three datasets in the timeline of year 2022.

1. **Data** - The data has information on car level. Like identification number, booking country (where the car was purchased), retail country (where the car will be refurbished), booking date (the date the car was booked), unbooking date (if the car was unbooked then the date of the event).
2. **Transport Data :**
   Dates : A transport have 3 relevant dates -
   1. Order date
   2. Delivery date
   3. Canceled date

   Type : transport status: =>
   1 = not yet ordered
   2 = ordered not delivered
   3 = delivered
   4 = canceled
3. **Daily Grid :**

Kpis definitions:

Booking : A car is selected for Autohero

First workshop entry : A car on its journey from booking to customer delivery can enter the workshop multiple times. The first time it enters the workshop is an event known as the ‘first workshop entry’.

Lead Time : Calendar days between two dates is a ‘lead time’. It could be order to delivery, booking to delivery, order to cancel or something similar.

Backlog: A car is in the workshop entry backlog if a the end of the week, it was actively booked and not yet delivered. As an example, if a car is booked on monday, first of March and delivered to the workshop on monday 15th of March. Then it’s in the backlog on week 1, with an age of 6 days, on week 2, with an age of 14 days

<u>Questions:</u>

1- Provide a query with the number of bookings per week per retail country

2- Provide a query with the first workshop entry per week per retail country with average associated lead time from booking to the first workshop entry

3- Provide a query for the computation per week per retail country of:

- number of bookings
- Deliveries to the first workshop
- Associated: lead time of those deliveries (from booking to first workshop entry)
- Backlog: cars that, at the end of a given week, were booked and not yet delivered to the first workshop
- age of backlog

4- Build an overview displaying those kpis per country per week properly formatted
You can add in here other kpis that you find interesting

Expected output:

- One file with the queries you wrote
- One file with the overview (either in Excel or in Google Sheets)

Example:
If a car is booked on monday 1st of January (week 1) and was delivered for the first time to a workshop on the 17th of January (week 3), then:

The car will count as:

- 1 booking for week 1
- 1 first workshop entry for week 3
- The associated leadtime of the first workshop entry would be 17 -1 = 16 days
- 1 for the backlog of week 1, with age: 7 - 1 = 6 (week one end on January 7th)
- 1 for the backlog of week 2, with age 14 -1 = 13 (week two end on January 14th)